Liu 2021 Multiple Myeloma — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Clonal MM plasma cells form a per-patient archipelago without integration (pre-Harmony iLISI 1.06). Judged on the regenerated embedding across a θ sweep (not a scalar): at θ=15 the 15 samples share ONE continuous manifold (island strip 42.5→28→11.7→2.4→2.7% as integration rises to a clean-speck level), well mixed per-neighbourhood (iLISI 2.27 of 5.62 ceiling; patient |r|≤0.19 on diffusion axes). Detached specks are unintegrated clones of the same states, not a distinct lineage.

Gate 2 — module geography → GO (non-branching)

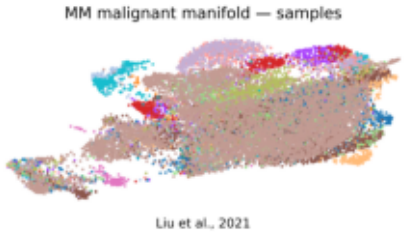
Single developmental axis: Gast & pSom vary along it with organised cross-repression (smoothed r=−0.67), and Crest occupies the opposite terminal end (pSom↔Crest r=−0.95). No Gast/pSom co-expression hub and no opposing Gast arm → non-branching. FLAG: Crest enriches the Gast end (Crest↔Gast r=+0.56, Crest↔pSom r=−0.95) — inversion of the canonical pSom-arm Crest tip; flagged but PASSES.

Gate 3 — hub cross-repression → NA

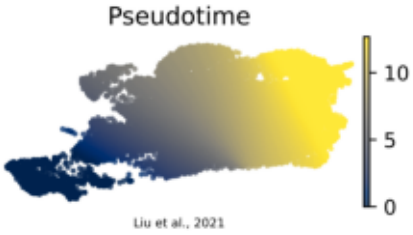
Non-branching manifold has no bifurcation; the tree-cluster cross-repression test does not apply.

STEP FIGURES

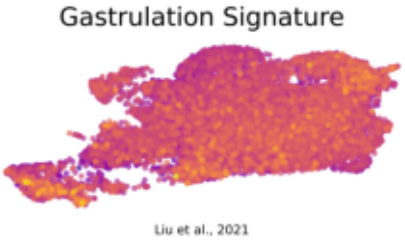
Manifold — samples



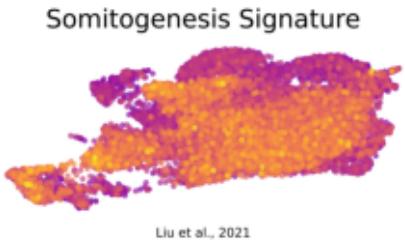
Pseudotime



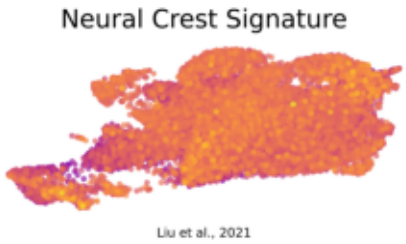
Gastrulation



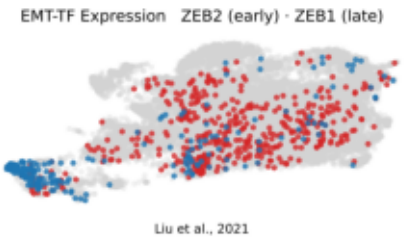
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple myeloma plasma cells (Liu et al., 2021), 10x — 24 samples / 74,986 cells (master scope); 17,267 malignant plasma cells; 15 marrows pass the 66→156 count cliff (≥100 cells; 17,003), 16,551 retained after removing detached islands. Global module means Gast −0.024, pSom +0.032, Crest −0.006 → Linear-pSom/Crest class (pSom the strongest module).

Verdict.
The embryonic-development pattern REPRODUCES as an organised non-branching cross-repression axis (Gast/pSom early, Crest terminal), with a Crest-on-Gast-arm inversion flag. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (early) → ZEB1 (late) — the only factors above the 3% detection floor.

Parameter choices

Min-cell threshold	≥100 cells/sample (drops 5 thin marrows; count cliff 66→156, ratio 2.4)	**Integration**	Harmony θ=15 on sample, 20 PCs (iLISI plateaus + island fraction bottoms out)
Embedding	PCA-50, kNN=30, UMAP spread 1.3	**Geometry**	Non-branching principal curve (ElPiGraph, fold-free SA 2.9; no hub/arm)
Scope	Whole manifold (no lobe excised); axis orthogonal to proliferation	**Root**	Gast/pSom-high, Crest-low tip (gastrulation/somitogenesis early, crest late)
EMT-TF panel	ZEB2 (early) & ZEB1 (late) clear the >3% floor; VIM excluded		